

Pradip Sapkota

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EDUCATION

Minnesota State University, Mankato

B.S. Computer Engineering, GPA: 3.7/4.0

Mankato, MN

Expected Dec. 2026

TECHNICAL SKILLS

Programming: C, C++, Python, MATLAB, SystemVerilog, VHDL, SQL, Ladder Logic

Hardware & Embedded: FPGA, Microcontrollers, Raspberry Pi, Sensors, Digital Logic, PLC, Modbus TCP

Engineering Tools: KiCad, Autodesk EAGLE, Oscilloscope, Spectrum Analyzer, AutoCAD, SolidWorks, ModelSim

Testing & Systems: Hardware Debugging, System Validation, Linux, Git/GitHub, AWS Lambda, DynamoDB

PROJECTS

Smart Door Access Control System | *KiCad, Autodesk EAGLE, Embedded Systems, Python*

Apr. 2025

- Designed and fabricated an access-control PCB using KiCad and Autodesk EAGLE to integrate locks, sensors, and authentication hardware.
- Implemented access-control logic with anti-spoofing and fail-safe behavior; performed reliability testing across hardware and software.

Interactive AI Voice Assistant | *KiCad, Autodesk EAGLE, Raspberry Pi, Python*

Dec. 2024

- Designed and fabricated a PCB using KiCad and Autodesk EAGLE for a Raspberry Pi voice assistant with audio and GPIO-controlled hardware.
- Integrated Python software, audio, GPIO control, and external AI services; debugged hardware-software interactions during system testing.

EXPERIENCE

Minnesota State University, Mankato

IT Solutions Consultant

Jan. 2024 – Present

Mankato, MN

- Diagnose hardware, software, network, authentication, and system-integration failures across major desktop and mobile platforms.
- Perform root-cause analysis to isolate device, connectivity, configuration, permission, and enterprise-application failures.
- Support deployment, configuration, validation, and documentation while resolving **50+ technical incidents**.

Ruppell Inc.

Engineering Associate – Software, Controls & Hardware

Jul. 2025 – Aug. 2026

Mankato, MN

- Contributed to PLC design and integrated-circuit selection for industrial systems; performed hardware functional testing.
- Developed sensor-monitoring dashboards with React and AWS; improved data-processing and API performance by **40%** using Python workflows and API optimization.
- Supported hardware-software integration and defect resolution through requirements review and functional testing.

Minnesota State University, Mankato

Undergraduate Research Assistant – Edge-AI Digital Twin

Aug. 2023 – Jan. 2024

Mankato, MN

- Supported digital-twin research integrating real-time sensor data, embedded computing, and cyber-physical system monitoring.
- Assisted with Python development, data processing, testing, and technical research for intelligent control experiments.
- Researched edge computing and predictive maintenance and contributed code, technical findings, and documentation.

LEADERSHIP

Institute of Electrical and Electronics Engineers (IEEE)

Events Lead

Aug. 2023 – Present

Mankato, MN

- Lead embedded-design competitions and technical workshops, coordinating schedules, equipment, logistics, and faculty/student teams.